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Statement of Objectives, Master's in Media Arts & Sciences | MIT Media Lab

I'm a Costa Rican scientist and inventor, and the focus of my career is to improve the accessibility of mental health therapies by developing technology that enables private user-driven delivery of psychological and psychiatric interventions. I would like to continue my work by pursuing a master's degree working with Dr. Cynthia Breazeal in the Personal Robots group, with Dr. Fadel Adib in the Signal Kinetics group, or with Dr. Pattie Maes in the Fluid Interfaces group.

Here in Latin America, as in many other regions of the world, mental health is still taboo. People often do not talk about therapy and deny any kind of psychological intervention due to mental health stigma, even if there's extensive evidence of its benefits. The culture of this region plays a role in shaping and enforcing the stigma, driven by two main factors: family and religion. Strong traditional family values often have a negative impact on seeking mental health support, especially when diagnosis could bring "shame on the family". In addition, religious beliefs also shape the stigma. Patients with chronic mental health conditions or their families often interpret the medical condition as a punishment for misbehavior. For this reason, people avoid talking about mental health. The taboo around mental health affects our society in multiple dimensions. For instance, a study published by the London School of Economics, indicates that the lost contribution to the economies of Latin America and the Caribbean, due to mental illness leading to disability or death among young people, is estimated at nearly $30.6 billion per year. In addition, according to the United Nations Children's Fund, in Latin America, more than 10 adolescents lose their lives every day due to suicide, and most of them never seek treatment.

Personally, I became aware of the magnitude of this problem through several years of volunteering supporting high school students from some of the most deprived neighborhoods in Costa Rica. There, I witnessed the impact of the mental health stigma when I realized that a large number of students suffer physical and physiological abuse at home, but usually avoid expressing their feelings about the abuse because talking about emotions is generally perceived as a weakness. As a consequence, they often build a strong personality to survive in their families, leaving no room for vulnerability. Extensive research has shown that an effective approach to alleviate this problem is enabling early access to mental-health support.

There is a pressing need to find new ways to support mental health in my region. Thus, my goal is to reduce the barrier caused by stigma, by increasing people's openness to psychological interventions through technology. The idea is to offer people the possibility to create spaces free of judgment, so they can explore and, eventually, change their misconceptions about mental health. Psychoeducational interventions and Cognitive Behavioral Therapy have proven successful to support various mental health conditions, and technology could provide access to them without feeling exposed to uncomfortable public situations. In this way, we will be reducing the barrier to the first contact with help, which eventually may facilitate receiving support from a professional.

My experience working at the Media Lab has inspired me to design new solutions to support people in identifying and avoiding behaviors that could affect their health. This is similar to "Saving Face", a project in which I participated, headed by Dr. Camilo Rojas, and advised by Dr. Pattie Maes, Dr. Fadel Adib, Dr. Kevin Esvelt, and Dr. Joseph A. Paradiso. We developed a system that leverages commercial earphones and smartphones to identify when the user is about to touch their face, and alert them in order to avoid the action. This results in a reduction in the spread of infections like influenza or COVID-19. My second project at the Media Lab, “Project Us” (which I have been co-leading with Dr. Camilo Rojas), seeks to teach people about emotions and how they can have an impact in their life. Recognizing emotions helps us to be in control of them and helps us to improve our reactions to certain situations. “Project Us” is a platform that uses artificial intelligence to share real-time feedback about users’ emotions. We have been using it to support individuals in measuring and improving inclusive behaviors.

Based on my goals and experiences until today, as described above, I strongly believe that my graduate studies at the Media Lab will enable me to enhance the effectiveness of psychological (health) interventions in three different ways.

First, Social Robots could help adults and teenagers to learn about and to practice mental wellbeing in a safe space. The teenagers I worked with avoid talking about those experiences because they feel ashamed of their feelings. Participating in a psychoeducational intervention guided by a robot, where they learn about emotion validation could help them feel safe and express their emotions. The robot will capture the attention of the user, enhancing the impact of the intervention, while reducing the pressure of feeling judged. The innovation will consist of providing more accurate and holistic information to the robot about the user's emotional state so that it can perform personalized interventions. According to state-of-the-art research in psychology, personalized interventions work better because each individual has various constellations of problems, and each problem can be targeted by a different solution. But one of the main challenges in testing is adapting each intervention because it is very time-consuming and requires a lot of information from the user. For instance, it is possible to use Jibo combined with high-frequency signals to measure breathing and heart rates. After some experiments with “Saving Face”, we came to the realization that it is possible to use these signals to monitor the user’s emotional state. The robot will be in charge of guiding a meditation and breathing session, to teach the user about relaxation and its benefits. We can use a combination of biosignals measured by the robot to provide a better intervention to the user by giving real-time feedback.

Regarding the Signal Kinetics group, I would like to create systems that leverage widely available commercial platforms (e.g., smartphones or laptops) to perform longitudinal monitoring of users’ emotions during their daily routine. Such systems would enhance the accuracy of diagnosis of mental health challenges and deliver just-in-time psychological interventions tailored for each user. The key difficulty to be addressed is to enable consumer devices to seamlessly track in real-time biosignals that contain information about emotions (e.g., heart rate, breathing rate, body movement patterns, etc.). I plan to begin tackling this challenge by implementing an ultrasound sonar system capable of monitoring the breathing and heart rate of a user from their own laptop. The base for this system will be the FMCW + Doppler signal processing pipeline that we developed for Saving Face, which will be enhanced by factoring in the phase information obtained from the FMCW spectrogram, time-multiplexing the transmission of ultrasound via each speaker of the laptop, and leveraging the spectrograms received in each microphone of the laptop. The result will be an improvement of the accuracy in biosignal measurement, plus enhanced robustness against confounding movements. An interesting demonstration of the system will be running a longitudinal test where the target biosignals of a subject will be monitored with the laptop across multiple days and with wearable devices (to provide a ground truth), while the tone of voice and facial expressions are monitored with the laptop. I expect this setup to exhibit correlations between emotional cues and the biosignals obtained through the wireless system. A future step could be studying the integration between ultrasound and RF signals (e.g., WiFi or Bluetooth), as a mechanism to further improve accuracy and robustness.

Working with the Fluid Interfaces group would enable me to explore the development of closed-loop systems to deliver self-guided psychological interventions. I'm particularly interested in studying the induction of trance-like states (e.g., through Hypnotherapy), and using this setup to design novel experiments oriented at better understanding the effects of these therapies in the body and mind. My first approach for this exploration will be to design an app that will deliver a guided meditation oriented at inducing Hypnosis, the app will also monitor the progress of the user in the hypnotic stage through tracking the verbal cues (leveraging a natural language processing engine for identifying the answers to standard monitoring questions) and measuring the respiration rate (using the ultrasound sonar that we developed for Saving Face). I will then proceed to run a user study to compare the effectiveness with open-loop commercial hypnosis systems (e.g., the hypnosis app for Alexa) and, hopefully, collaborate with a professional practitioner to compare the performance with the golden standard (i.e., the intervention delivered directly by a professional). I expect this exploration to open new venues for improving the accessibility of Hypnotherapy and develop a setup to better understand the effectiveness and principles of operation of this intervention.

Whether I use robots, wearables, wireless sensors, or a combination of these, my north is to help people to overcome the mental health stigma. However, it is important to remember that Latin America is a region with strong social inequalities. This means that not all people could have access to these solutions. Social entrepreneurship could help me transform my solution into a product that people could use outside the lab. By designing a business model that could help people while the products continue to improve it is possible to reduce the economic barrier for access. For this reason, I see the connection between industry and research the MIT Media offers as an opportunity to launch my own startup. The company that I have in mind will be commercializing tools based on my research made in the Media Lab. Giving access to this technology to Latin American people, we will be supporting the region to reduce the access barriers caused by the mental health stigma.

Portfolio link